

Notes: Berk and van Binsbergen (JFE, 2015)

Measuring Skill in the Mutual Fund Industry

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1 Big picture

- **Question.** How should we measure mutual fund manager skill, and what does the correct measure imply about whether skill exists in the mutual fund industry?
- **Setting.** The paper studies actively managed U.S. mutual funds and argues that the right measure of skill is not abnormal return per dollar, but the *dollar value* a fund extracts from capital markets.
- **Core challenge.** Standard measures—especially net alpha and gross alpha—confound managerial skill with equilibrium capital-market outcomes and fund scale. In equilibrium, skilled managers may not deliver positive net alpha to investors, and a small high-return fund can add less value than a huge lower-return fund.
- **Main methodological result.** The paper defines skill as *value added*: the fund’s gross excess return over its benchmark multiplied by lagged assets under management (AUM). This is a dollar measure, not a return measure.
- **Main empirical result.** The average actively managed mutual fund adds about \$3.2 million per year (time-weighted / ex post mean) and about \$1.68 million per year in the ex ante cross section (\$140,000 per month).
- **Persistence result.** Cross-sectional differences in value added persist out of sample for as long as **10 years**. This is the paper’s strongest evidence that skill exists.
- **Investor-learning result.** Investors appear able to identify skill. Funds with higher current compensation (AUM times fee) subsequently generate higher future value added.
- **Benchmarking result.** Using a feasible, tradable benchmark based on Vanguard index funds, average investor net alpha is close to zero rather than strongly negative.
- **Decomposition result.** Roughly half of active funds’ value added reflects diversification services and the other half reflects stock-picking / market-timing skill.
- **Economic takeaway.** Mutual fund skill exists, but competition largely passes it into *fund size and fees*, not into persistent positive abnormal returns for investors. Net alpha measures investor competition; value added measures managerial skill.

2 Data and measurement (key objects)

2.1 Observed data

- **Main source.** The core data are the CRSP survivorship-bias-free mutual fund database, spanning **January 1962 to March 2011**.
- **Supplementary source.** The authors buy and combine monthly Morningstar database updates from **January 1995 to March 2011**. A key advantage is that historical Morningstar updates are mechanically free of survivorship bias because older updates cannot be revised after delivery.
- **Additional validation.** The authors use Bloomberg to resolve some remaining return discrepancies and SEC EDGAR filings to recover missing expense ratios.

- **Main sample used for the Vanguard benchmark.** The benchmark-based analysis starts in **January 1977**, right after Vanguard introduced its S&P 500 index fund.
- **Unit of observation.** The fund, not the individual manager.
- **Frequency.** Monthly returns, fees, and assets under management.

2.2 Funds, benchmarks, and sample construction

- **Fund universe.** The paper studies *all actively managed equity mutual funds*, not only domestic-U.S.-equity funds.
- **Why this matters.** Restricting attention to purely domestic-stock funds discards an increasingly important part of the mutual fund industry and also throws away managers who move capital between U.S. and non-U.S. securities.
- **Funds excluded.**
 - index funds,
 - bond funds,
 - money market funds,
 - observations before inflation-adjusted AUM reaches \$5 million,
 - and funds with fewer than 2 years of data.
- **Share classes.** Different share classes are aggregated up to a single fund so subclasses do not artificially inflate statistical significance.
- **Final sample sizes.**
 - **5,974** funds in the Vanguard-benchmark sample,
 - **6,054** funds in the Fama–French–Carhart (FFC) risk-adjusted sample.
- **Benchmark portfolios.** The main benchmark set is built from **11 Vanguard index funds**, including S&P 500, Extended Market, Small-Cap, European, Pacific, Value, Balanced, Emerging Markets, Mid-Cap, Small-Cap Growth, and Small-Cap Value funds.

2.3 Key descriptive objects

- **Domestic-only funds are shrinking in importance.** The fraction of total mutual-fund AUM managed by funds that exclusively hold domestic stocks falls from about **45% in 1977** to about **23% in 2011**.
- **Typical fund size is stable.** Median real fund size stays roughly flat over the sample period; industry growth mainly comes from *more funds*, not much larger median funds.
- **Distribution of value added (Vanguard benchmark).**
 - Time-weighted mean: \$0.27 million per month.
 - Cross-sectional mean: \$0.14 million per month.
 - Median: about $-\$0.02$ million per month.

- 90th percentile: \$0.75 million per month.
- 95th percentile: \$1.80 million per month.
- 99th percentile: \$7.82 million per month.
- **Most funds destroy value individually.** About **57%** of funds have estimated average value added below zero, but because skilled managers run disproportionately more capital, the *industry-wide average* value added is positive.
- **Investor returns.** Average net alpha to investors is close to zero with the Vanguard benchmark:
 - equally weighted: **2.74 bp/month**,
 - value weighted: **−0.95 bp/month**,
neither significantly different from zero.

2.4 Unobservables, cleaning, and timing

- **Return cleaning.** About **3.3%** of return observations differ between CRSP and Morningstar, and even **1.3%** differ by more than 10 bp. The paper resolves many of these inconsistencies using NAV/dividend data and Bloomberg checks.
- **AUM cleaning.** Roughly **16%** of AUM observations differ across CRSP and Morningstar; missing AUM is filled using the most recent past observation after extensive cross-checking.
- **Expense ratios.** More than **40%** of CRSP expense-ratio observations are missing initially, but most are recovered using fiscal-year extension, Morningstar data, and SEC filings.
- **Benchmark timing issue.** Because Vanguard benchmark funds enter over time, the paper orthogonalizes the benchmark set and uses an augmented basis to estimate benchmark exposures consistently even when some benchmark funds do not yet exist.
- **Persistence-test timing.** To avoid spurious persistence from using the same beta estimates in both ranking and evaluation, betas for the measurement horizon are estimated only with data from that horizon, not with prior sorting data.

3 Models and empirical specifications (all equations + interpretation)

3.1 Return decomposition and traditional alpha measures

Let R_{it}^n denote the excess return earned by investors in fund i at time t . The paper decomposes this as

$$R_{it}^n = R_{it}^B + \varepsilon_{it},$$

where R_{it}^B is the return on the investor’s next-best feasible alternative investment (the benchmark), and ε_{it} is the deviation from that benchmark.

The **net alpha** of fund i is the time-average abnormal return earned *by investors*:

$$\hat{\alpha}_i^n = \frac{1}{T_i} \sum_{t=1}^{T_i} (R_{it}^n - R_{it}^B) = \frac{1}{T_i} \sum_{t=1}^{T_i} \varepsilon_{it}.$$

If $f_{i,t-1}$ is the fee charged between $t - 1$ and t , then the fund's **gross excess return** is

$$R_{it}^g = R_{it}^n + f_{i,t-1} = R_{it}^B + \varepsilon_{it} + f_{i,t-1},$$

and the **gross alpha** is

$$\hat{\alpha}_i^g = \frac{1}{T_i} \sum_{t=1}^{T_i} (R_{it}^g - R_{it}^B).$$

Interpretation. Net alpha is an investor-return object. Gross alpha is a return-before-fees object. Neither is naturally a *dollar-value* measure of managerial skill.

3.2 Value added as the paper's skill measure

Let $q_{i,t-1}$ be real AUM (inflation adjusted) at the end of the previous period. The realized **value added** between $t - 1$ and t is

$$V_{it} \equiv q_{i,t-1} (R_{it}^g - R_{it}^B) = q_{i,t-1} f_{i,t-1} + q_{i,t-1} \varepsilon_{it}.$$

Thus value added equals two parts:

- dollar fees collected by the fund, plus
- the benchmark-adjusted dollar value the fund creates for (or extracts from) investors.

The paper defines managerial skill as the expected value added:

$$S_i \equiv \mathbb{E}[V_{it}].$$

The fund-level estimator is

$$\hat{S}_i = \frac{1}{T_i} \sum_{t=1}^{T_i} V_{it}.$$

Two averages matter:

$$S = \frac{1}{N} \sum_{i=1}^N \hat{S}_i,$$

$$S^W = \frac{\sum_{i=1}^N T_i \hat{S}_i}{\sum_{i=1}^N T_i}.$$

The first is the mean of the *ex ante* distribution of skill; the second is the mean for surviving funds weighted by time in sample.

Interpretation. Value added measures the total dollar amount extracted from capital markets. It naturally accounts for scale, which return measures do not.

3.3 Null hypotheses and what they imply

The paper distinguishes two no-skill hypotheses.

Strong-form no-skill hypothesis:

$$S_i = 0 \quad \text{for every } i.$$

No manager has skill.

Weak-form no-skill hypothesis:

$$S = \frac{1}{N} \sum_{i=1}^N \hat{S}_i = 0.$$

Some managers could be skilled, but on average active management adds no value.

The alternative hypothesis is that managers can have skill and, at worst, unskilled managers can always index rather than deliberately destroy value. Therefore,

$$S_i \geq 0 \quad \text{for every } i,$$

with strict inequality for some skilled managers.

Interpretation. Under the strong null, value added should not be persistent across funds. Under the alternative, cross-sectional skill differences imply persistence.

3.4 Why net alpha and gross alpha fail to measure skill

The paper proves its core theoretical point with a decreasing-returns-to-scale formulation. Suppose manager i 's active-management alpha is

$$\alpha_i^n(q) = a_i - b_i q,$$

where $a_i > 0$ is the alpha on the first dollar actively managed and $b_i > 0$ captures decreasing returns to scale.

The manager chooses active capital q to maximize dollar value added:

$$V_i = \max_q q(a_i - b_i q).$$

The first-order condition gives

$$q_i^* = \frac{a_i}{2b_i}.$$

Substituting back,

$$V_i = q_i^*(a_i - b_i q_i^*) = \frac{a_i^2}{4b_i}.$$

The alpha earned on the *actively managed* part is then

$$\alpha_i^n = a_i - b_i q_i^* = \frac{a_i}{2}.$$

If total fund AUM is q_i , then the **gross alpha of the whole fund** is

$$\alpha_i^g = \frac{q_i^*}{q_i} \alpha_i^n = \frac{a_i^2}{4q_i b_i}.$$

Hence,

$$q_i \alpha_i^g = \frac{a_i^2}{4b_i} = V_i.$$

Implications (Proposition 1).

- Net alpha does *not* measure skill.
- Gross alpha measures skill only if all funds have identical AUM; it proxies for skill only if all funds are the same size.
- Value added, $q_i \alpha_i^g$, always measures skill.

Interpretation. The paper’s central insight is that skill is a *dollar object*. Gross alpha ignores scale; net alpha is driven to zero by investor competition in equilibrium.

3.5 Benchmark construction: Vanguard versus FFC

The paper uses two benchmark concepts.

1. Vanguard benchmark (preferred). The benchmark return is a projection of the active fund onto the set of tradable, marketed Vanguard index funds:

$$R_{it}^B = \sum_{j=1}^{n(t)} \beta_{ji} R_{jt},$$

where R_{jt} is the return on Vanguard index fund j and $n(t)$ is the number of such funds available at date t .

2. Fama–French–Carhart (FFC) risk adjustment. The benchmark is a portfolio of the four factor-mimicking portfolios:

$$R_{it}^B = \beta_i^{mkt} MKT_t + \beta_i^{smb} SMB_t + \beta_i^{hml} HML_t + \beta_i^{umd} UMD_t.$$

Interpretation. Vanguard funds are used as the *actual alternative investment opportunity set*. FFC is reported for completeness as a risk-adjustment device, but the paper argues it is not a valid ex ante benchmark because it embeds hindsight and, in some cases, nontradable strategies.

3.6 Persistence tests and order-statistic inference

To rank managers, the paper builds a **skill ratio**

$$SKR_i^\tau = \frac{\hat{S}_i^\tau}{\sigma(\hat{S}_i^\tau)},$$

where $\hat{S}_i^\tau$ is the estimated average value added up to month τ and $\sigma(\hat{S}_i^\tau)$ is its estimated standard deviation.

Funds are sorted into deciles using the skill ratio after at least three years of history. Then future value added is measured over horizons from **3 to 10 years**.

A central implementation issue is beta-estimation noise. If the same beta estimates are used in both the ranking period and the evaluation period, spurious persistence could arise. The paper therefore estimates benchmark exposures using *only* the measurement horizon. For the Vanguard benchmark, the first **18 months** of the evaluation window are dropped; for FFC, the first **6 months** are dropped.

The main nonparametric test counts:

- how often the top decile beats the bottom decile, and
- how often the top decile lies in the top half of funds ex post.

Under the strong no-skill null, these are exact **Binomial**($n, 1/2$) statistics.

Interpretation. This gives a distribution-free way to assess persistence that is robust to cross-fund correlation and excess kurtosis in value added.

4 Identification and instruments (what makes the estimates credible)

4.1 What fails in the standard approach

- **Net alpha fails because it is an equilibrium object.** In competitive capital markets with rational investors, nonzero net alpha should be competed away.
- **Gross alpha fails because it ignores scale.** A small fund with a high return can add less dollar value than a large fund with a modest return.
- **Sharpe's arithmetic is not enough.** Even if active investors as a group hold the market portfolio, that does not rule out mutual fund skill, because active managers may earn rents from other active investors or from liquidity traders.

4.2 What identifies skill here

- **Dollar value added relative to a feasible benchmark.** Skill is identified from the product of abnormal gross return and lagged AUM.
- **Cross-sectional persistence.** If skill is real, better funds should continue to add more value in the future. The paper shows this persistence up to 10 years.
- **Compensation predicts future performance.** If investors can infer skill, current AUM $\times$ fee should forecast future value added. It does.

4.3 Why the Vanguard benchmark is especially important

- It is composed of funds that were actually *tradable and marketed at the time*.
- It includes diversification services and transaction costs faced by passive investors.
- It avoids hindsight bias embedded in using factor portfolios as if they were always-investable benchmarks.
- Table 2 shows exactly why this matters: relative to Vanguard benchmarks, the FFC portfolios have positive alphas of about **22 bp/month** for SMB, **35 bp/month** for HML, and **70 bp/month** for UMD. That means the factors were better ex post opportunities than what passive investors could actually access in real time.

4.4 Why the empirical application is credible

- The CRSP sample is survivorship-bias free, and Morningstar monthly vintages preserve what investors could actually observe in real time.
- The authors carry out extensive data cleaning for returns, fees, AUM, and passive-fund classification.
- The sample includes international-equity exposure rather than artificially restricting attention to purely domestic funds.
- The persistence tests explicitly avoid look-ahead bias and mechanical beta-estimation persistence.

4.5 Relation to the literature

- The paper reinterprets the classic mutual-fund evidence: zero net alpha is not proof of zero skill.
- It is closely connected to Berk and Green (2004): competition drives investor alpha toward zero while scale allocates more capital to skilled managers.
- It also speaks to the efficient-market literature by separating the question “Are investors earning abnormal returns?” from “Do managers create rents?”

5 Tables and figures

5.1 Table 1 and Table 2: benchmark construction and why the FFC benchmark is problematic

Table 1

Benchmark Vanguard index funds. This table lists the set of Vanguard index funds used to calculate the Vanguard benchmark. The listed ticker is for the investor class shares which we use until Vanguard introduced an Admiral class for the fund, and thereafter we use the return on the Admiral class shares (Admiral class shares have lower fees but require a higher minimum investment).

Fund name	Ticker	Asset class	Inception date
S&P 500 Index	VFINX	Large-Cap Blend	08/31/1976
Extended Market Index	VEXMX	Mid-Cap Blend	12/31/1987
Small-Cap Index	NAESX	Small-Cap Blend	01/01/1990*
European Stock Index	VEURX	International	06/18/1990
Pacific Stock Index	VPACX	International	06/18/1990
Value Index	VVIAX	Large-Cap Value	11/02/1992
Balanced Index	VBINX	Balanced	11/02/1992
Emerging Markets Stock Index	VEIEX	International	05/04/1994
Mid-Cap Index	VIMSX	Mid-Cap Blend	05/21/1998
Small-Cap Growth Index	VISGX	Small-Cap Growth	05/21/1998
Small-Cap Value Index	VISVX	Small-Cap Value	05/21/1998

* NAESX was introduced earlier but was originally not an index fund. It was converted to an index fund in late 1989, so the date in the table reflects the first date we included the fund in the benchmark set.

Table 2

Net alpha of Fama-French-Carhart (FFC) portfolios. We regress each FFC factor portfolio on the Vanguard benchmark portfolios. The table lists the estimate (in bp/month) and t -statistic of the constant term (alpha) of each regression, as well as the R^2 of each regression.

	MKT	SMB	HML	UMD
Alpha (bp/month)	2	22	35	70
t -statistic	0.83	2.80	3.37	3.38
Adjusted R^2	99%	74%	52%	15%

Read:

- Table 1 lists the 11 Vanguard index funds used to build the feasible benchmark set. The key point is that the investable passive opportunity set expands through time, starting with the S&P 500 fund in 1976 and widening in the 1990s.
- Table 2 regresses the FFC factor portfolios on the Vanguard benchmark set. The market factor has essentially zero alpha, but SMB, HML, and UMD all have positive alphas relative to Vanguard benchmarks.
- This is the most transparent evidence that FFC factors are *not* appropriate ex ante benchmark portfolios. They contain hindsight and, especially for momentum, ignore implementation costs.

5.2 Figure 1 and Figure 2: the mutual fund universe changed, but typical fund size did not

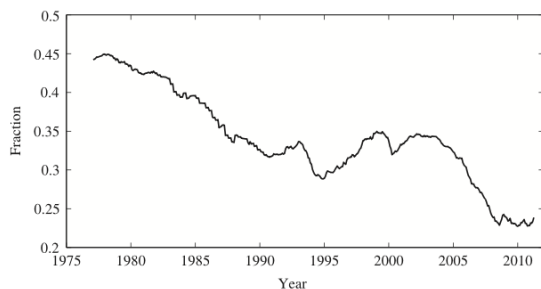


Fig. 1. Total assets under management (AUM) of domestic funds as a fraction of the AUM of all funds. The graph displays the AUM of mutual funds that exclusively hold domestic stocks as a fraction of the total AUM of all mutual funds, including those that invest in international stocks.

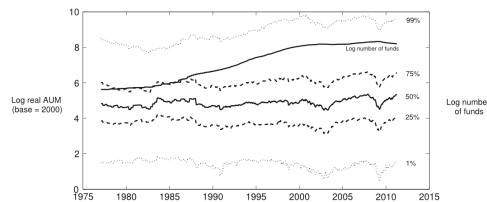


Fig. 2. Fund size distribution. The graph displays the evolution of the distribution of the logarithm of real assets under management (AUM) in millions of dollars (base year is 2000) by plotting the 1st, 25th, 50th, 75th, and 99th percentiles of the distribution at each point in time. The smooth black line is the logarithm of the total number of funds (not assets under management).

Read:

- Figure 1 shows that domestic-only equity funds became a much smaller share of the industry: from about 45% of AUM in the late 1970s to about 23% by 2011.
- So restricting the sample to domestic-only funds discards an increasingly important part of the active mutual fund industry.
- Figure 2 shows that the median real size of funds stayed roughly constant. Industry growth came primarily through a greater number of funds, not a dramatic increase in the size of the representative fund.
- This matters because it supports the stationarity of the product “abnormal gross return $\times$ lagged AUM,” the object used to measure value added.

5.3 Table 3: average value added and its cross-sectional distribution

Read:

- This is the headline table. Using the Vanguard benchmark, the ex post weighted mean value added is **\$0.27 million per month**, or about **\$3.2 million per year**.
- The ex ante cross-sectional mean is **\$0.14 million per month**.

Table 3
Value added ($\hat{S}_t$). For every fund in our database, we estimate the average monthly value added, $\hat{S}_t$. The cross-sectional mean, standard error of mean, t-statistic and percentiles are the statistical properties of this distribution. Percent with less than zero is the fraction of the distribution that has value added estimates less than zero. The cross-sectional weighted mean, standard error of the weighted mean and t-statistic are computed by weighting by the number of periods the fund exists, that is, they are the statistical properties of S_w defined by Eq. (9). The numbers are reported in Y2000 \$ millions per month.

	Vanguard benchmark	FFC risk measure
Cross-sectional weighted mean	0.27	0.25
Standard error of the weighted mean	0.05	0.06
t-statistic	5.74	3.94
Cross-sectional mean	0.14	0.10
Standard error of the mean	0.03	0.03
t-statistic	4.57	3.43
1st percentile	-3.60	-3.93
5th percentile	-1.15	-1.43
10th percentile	-0.59	-0.77
50th percentile	-0.02	-0.03
90th percentile	0.75	0.70
95th percentile	1.80	1.98
99th percentile	7.82	6.76
Percent with less than zero	57.01%	59.70%
Total number of funds	5974	6054

- The distribution is wide: the median fund slightly destroys value, but the upper tail is enormous, with the 99th percentile around **\$7.82 million per month**.
- About **57%** of funds have negative estimated value added, but skilled managers control enough capital that the industry average remains positive.

5.4 Figure 3 and Table 4: persistence of value added

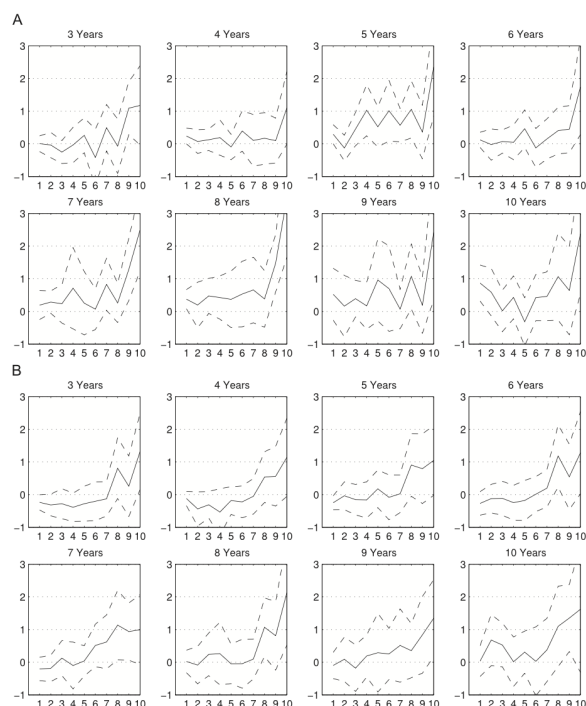


Fig. 3. Out-of-sample value added. Each graph displays average out-of-sample value added, $\bar{S}_t$ (in millions of 2000 dollars/month), of funds sorted into deciles on the skill ratio (horizontal axes), over the future horizon indicated by the graph title. The solid line indicates the performance of each decile and the dashed lines indicate the two standard error bounds. Panel A shows the results when value added is computed using Vanguard index funds as benchmark portfolios and Panel B shows the results using the Fama-French-Carhart (FFC) risk adjustment.

Table 4
Out-of-sample performance of the top decile. The table reports the average value added of the top decile at each horizon and the associated p-value, the fraction of time and associated p-value that the top decile has a higher value added realization than the bottom decile, the fraction of time the realized value added of the top decile is in the top half, and the average fraction of total AUM in the top decile. All p-values are one tailed; that is, they represent the probability, under the null hypothesis, of the observed test-statistic value or greater.

Horizon	Value added		Top outperforms bottom		Top in top half		Fraction of total AUM (%)
Years	\$ Mil	p-value (%)	Freq. (%)	p-value (%)	Freq. (%)	p-value (%)	
Panel A: Vanguard benchmark							
3	1.19	2.51	56.32	4.75	56.32	4.75	24.82
4	1.10	2.49	57.14	2.07	59.45	0.32	25.56
5	2.32	0.11	55.81	3.54	56.98	1.46	24.34
6	1.72	0.95	57.09	1.09	57.46	0.79	25.30
7	2.47	0.00	61.57	0.01	64.55	0.00	22.57
8	3.44	0.01	58.23	0.67	58.65	0.46	25.65
9	2.42	1.00	54.23	9.15	55.31	4.50	24.94
10	2.38	0.52	54.69	5.55	57.93	0.31	24.95
Panel B: FFC risk adjustment							
3	1.30	1.33	56.13	0.47	57.63	0.06	17.93
4	1.13	3.01	58.34	0.02	57.72	0.05	19.50
5	1.03	2.68	59.60	0.00	58.79	0.01	17.88
6	1.27	2.22	58.85	0.01	56.50	0.28	19.38
7	0.98	3.37	59.71	0.00	56.12	0.44	17.91
8	2.13	0.42	59.12	0.01	57.14	0.13	19.01
9	1.35	1.12	56.51	0.18	55.15	1.09	16.10
10	1.62	4.67	58.91	0.01	56.74	0.22	21.83

Read:

- Figure 3 sorts funds by the skill ratio and then tracks future value added over 3- to 10-year horizons. The top decile always lies above the bottom decile, and the top-decile mean is positive at every horizon.
- Table 4 formalizes this. The top decile adds roughly **\$1.1–\$3.4 million per month** out of sample depending on the horizon.
- The exact Binomial order-statistic tests reject the no-skill null at almost every horizon with the Vanguard benchmark, and even more strongly with FFC risk adjustment.
- The last column shows that the top decile controls roughly **one quarter of total AUM**. This is consistent with investors directing capital toward managers who previously revealed skill.

5.5 Figure 4, Figure 5, and Table 5: investors appear able to identify and reward skill

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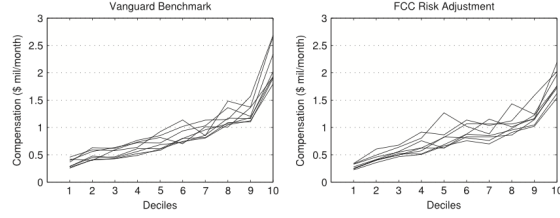


Fig. 4. Out-of-sample compensation. The plots display the average out-of-sample monthly compensation of each decile sorted on skill ratio using the Vanguard benchmark and the Fama-French-Carhart (FCC) risk adjustment. Each line in the plots represents a different horizon, which varies between three and ten years. For ease of comparison, the data sample (time period) is the same for both plots.

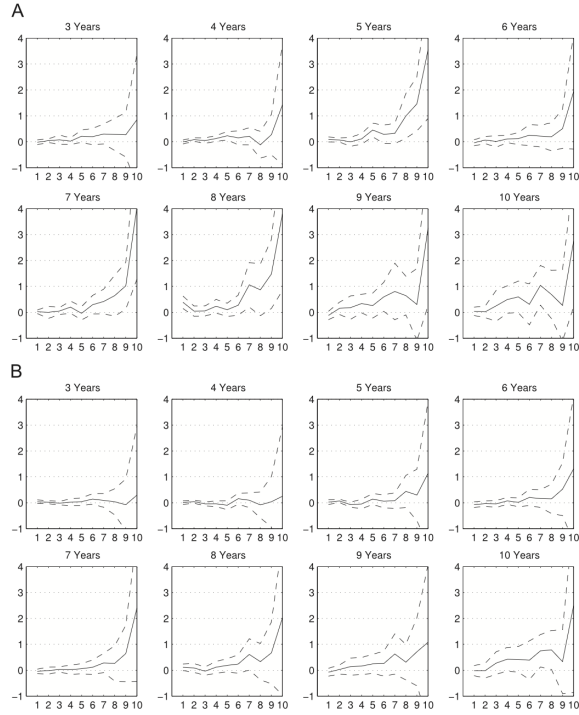


Fig. 5. Value added sorted on compensation. Each graph displays average out-of-sample value added, $\hat{S}_i$ (in millions of Y2000 dollars/month), of funds sorted into deciles based on total compensation (fres > assets under management). The solid line indicates the performance of each decile, and the dashed lines indicated the 95% confidence bands (two standard errors from the estimate). Panel A shows the results when value added is computed using Vanguard index funds as benchmark portfolios, and Panel B shows the results using the Fama-French-Carhart (FCC) risk adjustment.

Table 5
Out-of-sample monotonicity. At each horizon, we calculate the number of times each decile outperforms the next lowest decile. The table shows the p-value (in percent) of the observed frequency under the null hypothesis that skill does not exist, i.e., for a sample length of N months, the probability of the event is $\text{Binomial}(9, N/2)$.

Horizon	Vanguard benchmark Skill ratio	Compensation	FCC risk adjustment Skill ratio	Compensation
3	18.55	9.17	2.76	72.15
4	3.51	1.87	12.20	4.89
5	18.61	0.02	8.40	48.81
6	4.57	9.23	0.58	3.24
7	15.92	5.85	0.53	17.52
8	4.16	3.79	4.88	25.58
9	28.61	14.71	5.97	3.90
10	53.02	0.25	16.38	4.82

- Figure 4 shows that funds sorted into higher skill-ratio deciles earn higher future compensation ($AUM \times \text{fee}$).
- Figure 5 flips the exercise: sort funds by current compensation. The result is slightly *stronger* monotonicity in future value added than when sorting by the skill ratio.
- Table 5 confirms this statistically: compensation sorts produce lower p-values and cleaner monotonicity than skill-ratio sorts, especially with the Vanguard benchmark.
- The paper’s interpretation is striking: investors are not simply “dumb money” chasing past returns. They appear able to infer skill and allocate capital toward it.

5.6 Table 6: the returns to investors are near zero

Read:

- With the Vanguard benchmark, the equally weighted net alpha is about **2.74 bp/month** and the value-weighted net alpha is about **−0.95 bp/month**; neither is statistically different from zero.
- This is exactly what one would expect if investors compete away most of the rents from skill.
- With FFC risk adjustment, net alpha becomes negative, but the paper interprets that as evidence that FFC is not the relevant benchmark opportunity set rather than as proof that managers destroy value.

Table 6
Net alpha (in bp/month): The table reports the net alpha of two investment strategies: Investing \$1 every month by equally weighting over all existing funds and investing \$1 every month by value weighting (based on AUM) over all existing funds.

	Vanguard benchmark	FFC risk measure
Equally weighted	2.74	−3.88
t-statistic	0.73	−1.40
Value weighted	−0.95	−5.88
t-statistic	−0.31	−2.35

Table 7
Performance of active funds and index funds. The table computes the value added, net alphas and the p-value of the persistence order statistic that counts the number of times the top decile outperforms the bottom decile for the set of active mutual funds and compares it with the set of index funds (including the Vanguard index funds themselves). To separate the value added coming from diversification benefits versus stock picking or market timing, we use two different benchmarks: Vanguard index funds’ gross returns and Vanguard index funds’ net returns.

Benchmark	Active funds Vanguard gross	Vanguard net	Index funds Vanguard gross	Vanguard net
Number of funds	5974	5974	644	644
In sample value added				
Weighted mean (\$mil/mon)	0.16	0.27	−0.02	0.11
t-statistic	3.46	5.74	−0.20	0.94
Mean (\$mil/mon)	0.07	0.14	−0.05	0.03
t-statistic	2.49	4.57	−0.49	0.37
1st percentile	−3.83	−3.60	−6.18	−5.98
5th percentile	−1.27	−1.15	−1.20	−1.15
10th percentile	−0.64	−0.59	−0.53	−0.47
50th percentile	−0.02	−0.02	−0.01	0.00
90th percentile	0.61	0.75	0.50	0.76
95th percentile	1.55	1.80	1.56	1.82
99th percentile	7.56	7.82	4.83	4.83
In-sample net alpha				
Equally weighted (bp/mon)	−	2.7	−	−1.9
t-statistic	−	0.73	−	−0.41
Value Weighted (bp/mon)	−	−1.0	−	1.3
t-statistic	−	−0.31	−	0.65
Persistence(p-value (% of the top decile outperforming the bottom decile at each horizon)				
3 year horizon	3.47	4.75	−	−
4 year horizon	3.87	2.07	−	−
5 year horizon	11.84	3.54	−	−
6 year horizon	9.23	1.09	−	−
7 year horizon	0.00	0.01	−	−
8 year horizon	1.87	0.67	−	−
9 year horizon	23.39	9.15	−	−
10 year horizon	18.14	5.55	−	−

5.7 Table 7: diversification services versus stock-picking and market-timing skill

Read:

- Switching the benchmark from Vanguard *net* returns to Vanguard *gross* returns strips out the value of diversification services provided by active funds.
- The value-weighted mean value added falls from **\$0.27 million/month** to **\$0.16 million/month**. So roughly 40% of total value added reflects diversification services and roughly 60% reflects stock selection / market timing.

Table 8

Gross alpha $\rho(S_t, \alpha_t^g)$ is the cross-sectional correlation of estimated gross alpha and average value added. R^2 is the fraction of the cross-sectional variation in S_t explained by α_t^g . The rest of the table reports the average in-sample gross alpha as well as the results of the out-of-sample persistence test using the frequency with which the tenth decile beats the first decile. We follow exactly the same procedure we used in the original persistence tests except instead of using value added, we use the gross alpha to construct the skill ratio and we assess future performance using the value weighted gross alpha of the decile.

	Vanguard benchmark	FCC risk adjustment
Explanatory power of gross alpha		
$\rho(S_t, \alpha_t^g)$	0.23	0.24
R^2	0.05	0.06
In sample gross alpha		
Value weighted mean (bp/month)	6.5 ^a	0.66
Equally weighted mean (bp/month)	13 ^b	5
Persistence(p-value (%) of top decile outperforming bottom)		
3 year horizon	17.28	0.79
4 year horizon	34.19	1.36
5 year horizon	1.40	0.00
6 year horizon	2.67	0.06
7 year horizon	29.13	0.33
8 year horizon	18.16	0.32
9 year horizon	31.42	1.09
10 year horizon	59.00	6.19

^a t-statistic greater (in absolute value) than 1.96

^b t-statistic greater (in absolute value) than 2.54.

- Persistence remains even after stripping out diversification services. This means the paper's main skill result is not only about passive diversification.
- The table also shows that relative to Vanguard gross returns, the average index fund does not beat Vanguard. So Vanguard appears unusually efficient as a diversification provider.

5.8 Table 8: gross alpha is a poor proxy for skill

Read:

- The cross-sectional correlation between value added and gross alpha is only about **0.23**, and gross alpha explains only about **5%** of the cross-sectional variation in skill.
- Average value-weighted gross alpha is about **6.5 bp/month**, which is basically the fee active managers charge.
- Using past gross alpha to sort managers yields little persistence with the Vanguard benchmark.
- This is the core empirical confirmation of Proposition 1: in the real world, fund size varies much more than fees, so gross alpha is a poor proxy for skill.

6 Short summary

- **Method.** The paper measures mutual fund skill by *value added*, the benchmark-adjusted gross return times lagged AUM, rather than by net or gross alpha.
- **Identification.** Skill is identified from dollar abnormal performance relative to a feasible passive benchmark and from its persistence out of sample. Exact order-statistic tests avoid relying on fragile t-statistics.
- **Application.** In a large cleaned sample of active equity mutual funds, average value added is positive and economically large, even though investor net alpha is near zero.
- **Main result.** Skill exists and is persistent for up to 10 years; investors appear able to identify it and reward it through fund flows and higher compensation. Gross alpha is a poor measure of that skill.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

- The paper establishes that the right way to measure mutual fund skill is in dollars, not in return percentages.
- Once skill is measured as value added, the evidence points strongly toward the existence of economically meaningful managerial skill.
- The paper also shows that competition in capital markets and the existence of skill are not contradictory: managers can be skilled while investors still earn near-zero net alpha.

7.2 Economic intuition

- A skilled manager's scarce resource is not a percentage return but the ability to profitably deploy a limited amount of active capital.
- Competition by investors pushes net abnormal returns toward zero, so the rents from skill are capitalized mainly into *scale*: better managers run bigger funds and earn higher total fees.
- Because fund size varies much more than fees, cross-sectional differences in skill mostly show up in dollar compensation and AUM, not in gross alpha.

7.3 Takeaway

This paper is important because it changes the object we should look at when asking whether active mutual fund management is skilled. The paper's answer is yes: skill exists, it is persistent, and investors seem able to find it. But the manifestation of skill is not positive alpha to investors. It is value added in dollars, together with the larger scale and compensation that skilled managers command.